

SOFTWARE TEST PLAN

Student Record System

1. Introduction

1.1 Purpose

The purpose of testing is to verify that the Student Record System satisfies the functional, non-functional, technical and security requirements defined in the updated SRS and behaves correctly under normal, invalid and unauthorized conditions.

1.2 Scope

Testing covers authentication, role identification and access control; student CRUD operations; grades; attendance; search and filtering; input validation; persistent storage; safe exit; error handling; usability; performance; data integrity; backup and recovery; reliability; portability; maintainability; UI consistency; modular C/C++ architecture; and the two security requirements SR-01 and SR-02.

1.3 Test Objectives

- Verify that the functional requirements FR-01 to FR-25 are satisfied.
- Verify that the non-functional requirements NFR-01 to NFR-12 are satisfied or evaluated through the appropriate test/review activity.
- Verify that the technical requirements MTR-01 to MTR-04 are satisfied through implementation and architecture/design review.
- Verify that SR-01 and SR-02 are satisfied through security validation.
- Verify that invalid, incomplete, duplicate and unauthorized inputs are handled correctly.
- Verify that data is stored, retrieved and preserved as required.
- Record actual results and update planned test status to Pass/Fail after execution.

1.4 Test Levels

ID Prefix	Test Level	Purpose
UT	Unit Test	Tests one function, class or module in isolation.
FT	Functional Test	Checks whether an implemented function satisfies its requirement.
ST	System Test	Checks the integrated Student Record System end-to-end.
AT	Acceptance Test	Checks whether the system satisfies the expected user/project requirement.
REV	Review	Checks architecture, modularity, portability, maintainability and other requirements that are primarily verified by inspection/review.

2. Test Strategy

Requirement-based testing will be used. Each test case below is explicitly mapped to the requirement IDs in the updated SRS. Positive and negative scenarios are included. Where a requirement is better verified by inspection or architecture review, the Test Plan identifies the review activity rather than inventing an execution scenario.

2.1 Functional Testing

Functional testing covers FR-01 to FR-25: login, authentication, role identification, role-based menus, student management, grades, attendance, search/filter, validation, persistent storage and safe exit.

2.2 Non-Functional Testing

Non-functional testing/review covers NFR-01 to NFR-12, including usability, performance, data integrity, error handling, maintainability, backup/recovery, reliability, access control, portability, UI consistency and storage/performance.

2.3 Security Testing

Security validation covers the two security requirements in the updated SRS: SR-01 authentication using valid credentials and SR-02 role-based restriction of student information and system operations. The SRS currently defines exactly two security objectives and two security requirements; the Test Plan therefore does not use SR-03 or SR-04.

3. Test Environment

Item	Planned Environment
Programming Language	C / C++
Compiler	Standard C/C++ compiler available on the target computer
Storage	Persistent storage selected by the project, such as text, CSV, binary file or permitted local database
Execution	Local desktop/terminal environment
Test Data	Student records containing Student ID, name, contact details, course/department, semester, subject/marks/grades and attendance
Scope	Core mini-project functionality and specified security controls

4. Test Cases

The following test cases are aligned with the requirement IDs and Test Case ID convention in the SRS. The SRS lists TC-FR-01 through TC-FR-25, TC-NFR-01 through TC-NFR-12, TC-MTR-01 through TC-MTR-04, and security traceability for SR-01 and SR-02. Test Plan uses the same IDs.

Test ID	SRS Req.	Test Scenario	Input / Condition	Expected Result
TC-FR-01	FR-01	Login with valid credentials	Valid Administrator, Faculty or Student credentials	Correct actor is allowed to log in.
TC-FR-02	FR-02	Authenticate credentials before access	Valid and invalid username/password combinations	Valid credentials grant access; invalid credentials are rejected.
TC-FR-03	FR-03	Verify role identification and authorization	Login separately as Administrator, Faculty and Student	Correct role is identified and only permitted functions are available.
TC-FR-04	FR-04	Verify role-based main menu	Login using each actor role	Menu displays only operations permitted for that role.
TC-FR-05	FR-05	Add a new student	Unique Student ID plus valid name, contact, course/department and semester	Student record is added successfully.
TC-FR-06	FR-06	Reject duplicate Student ID	Enter an existing Student ID	System rejects the new record and displays a duplicate-ID message.
TC-FR-07	FR-07	View student details according to role	Administrator/Faculty view a student; Student views own record	Permitted record is displayed; Student cannot view another student's record.
TC-FR-08	FR-08	View stored records according to access permissions	Administrator/Faculty request stored records	Records are displayed according to permissions.
TC-FR-09	FR-09	Search student by ID or name	Valid Student ID and valid student name searches	Matching student record(s) are displayed.
TC-FR-10	FR-10	Update student information	Existing Student ID and valid updated details	Administrator can update and save the record.
TC-FR-11	FR-11	Delete student after confirmation	Existing Student ID; confirm and cancel scenarios	Record is deleted only after confirmation; cancellation retains the record.
TC-FR-12	FR-12	Record subject marks/grades	Valid student, subject and marks	Faculty can record marks/grades.
TC-FR-13	FR-13	Update subject marks/grades	Existing grade and valid updated marks	Faculty can update the grade.
TC-FR-14	FR-14	View grades by permitted actor	Faculty views grades; Student views own grades	Correct grades are displayed according to role.
TC-FR-15	FR-15	Record attendance	Valid student and attendance value	Faculty can record attendance.
TC-FR-16	FR-16	Update attendance	Existing attendance record and valid new	Faculty can update attendance.

			value	
TC-FR-17	FR-17	View attendance by permitted actor	Faculty views attendance; Student views own attendance	Correct attendance is displayed according to role.
TC-FR-18	FR-18	Filter student records	Course, department or semester filter	Administrator/Faculty receive records matching the selected filter.
TC-FR-19	FR-19	Display search/filter results according to permissions	Search/filter using permitted and non-permitted actor roles	Only matching records allowed for the actor are displayed.
TC-FR-20	FR-20	Validate required fields and invalid data	Empty, incomplete or invalid required fields	Invalid record is rejected before saving.
TC-FR-21	FR-21	Re-enter invalid input	Enter invalid input followed by valid input	System prompts for correction and accepts valid re-entry.
TC-FR-22	FR-22	Save newly added or modified records	Add/update a valid student record	Changes are written to persistent storage.
TC-FR-23	FR-23	Load records at application start	Start application after records have been saved	Stored records are retrieved.
TC-FR-24	FR-24	Preserve records after restart	Save records, close application, restart	Previously saved records remain available.
TC-FR-25	FR-25	Safe exit from main menu	Administrator, Faculty and Student select Exit	Application exits safely without unintended data loss.
Test ID	SRS Req.	Test Scenario	Input / Condition	Expected Result
TC-NFR-01	NFR-01	Usability / menu-driven interface review	Run normal user workflows	Menu is simple and user-friendly for intended actors.
TC-NFR-02	NFR-02	Performance test	Up to 1,000 records; add/search/update/view	Normal operations complete within 2 seconds.
TC-NFR-03	NFR-03	Data integrity test	Attempt duplicate ID and inconsistent/invalid record changes	Duplicate IDs are prevented and stored records remain consistent.
TC-NFR-04	NFR-04	Error handling test	Invalid input and recoverable runtime-error conditions	System handles errors without unexpected termination.
TC-NFR-05	NFR-05	Maintainability/modularity review	Inspect C/C++ source structure and functions/classes	Code is organized into clearly defined modular files/functions/classes.
TC-NFR-06	NFR-06	Backup and recovery test	Create backup using selected storage mechanism and restore it	Stored student data can be backed up and recovered as implemented.
TC-NFR-07	NFR-07	Reliability/persistence test	Perform normal operations, save data, close/restart	Records remain consistent and saved data is not unintentionally lost.
TC-NFR-08	NFR-08	Unauthorized modification test	Attempt record modification using a role without permission	Unauthorized modification is prevented.
TC-NFR-09	NFR-09	Portability test	Build/run using a standard C/C++ compiler on target environment	Application runs in the target environment as specified.
TC-NFR-10	NFR-10	UI consistency test	Use all major student record operations	Prompts, menus and messages remain consistent.
TC-NFR-11	NFR-11	Increasing-records performance/storage test	Increase number of records within intended scope	Storage/retrieval continues with acceptable performance.
TC-NFR-12	NFR-12	Role-based information/function restriction test	Attempt access to functions/information from each role	Access is restricted according to authenticated role.
Test ID	SRS Req.	Verification	Evidence / Condition	Expected Result
TC-MTR-01	MTR-01	C/C++ modular architecture review	Inspect authentication, student, academic and attendance modules	Implementation uses C/C++ and modular functions/classes as required.
TC-MTR-02	MTR-02	Persistent storage design review	Inspect selected storage implementation	System uses permitted persistent storage for student records.
TC-MTR-03	MTR-03	Student ID data model review	Inspect student data structure and storage	Student ID is used as the unique primary identifier.
TC-MTR-04	MTR-04	Separation-of-concerns review	Inspect authentication, validation, record management and storage modules	Required operations are separated into reusable modules/functions.

5. Security Validation

The SRS defines two security objectives and two security requirements. Security validation is therefore mapped only to SR-01 and SR-02.

5.1 Security Validation Test Cases

Security Test ID	SRS Security Req.	Validation Test	Input / Condition	Expected Result
SEC-01	SR-01	Attempt login with valid and invalid credentials	Valid and invalid credentials	Only valid credentials authenticate the user.
SEC-02	SR-02	Attempt permitted and restricted operations using each actor role	Administrator, Faculty and Student sessions	Functions and student information are accessible only according to the authenticated role.

5.2 Test Execution Status

All tests are initially Planned after implementation results will be updated.

Test ID	Status	Actual Result	Remarks
TC-FR-01	Planned		

TC-FR-02	Planned		
TC-FR-03	Planned		
TC-FR-04	Planned		
TC-FR-05	Planned		
TC-FR-06	Planned		
TC-FR-07	Planned		
TC-FR-08	Planned		
TC-FR-09	Planned		
TC-FR-10	Planned		
TC-FR-11	Planned		
TC-FR-12	Planned		
TC-FR-13	Planned		
TC-FR-14	Planned		
TC-FR-15	Planned		
TC-FR-16	Planned		
TC-FR-17	Planned		
TC-FR-18	Planned		
TC-FR-19	Planned		
TC-FR-20	Planned		
TC-FR-21	Planned		
TC-FR-22	Planned		
TC-FR-23	Planned		
TC-FR-24	Planned		
TC-FR-25	Planned		
TC-NFR-01	Planned		
TC-NFR-02	Planned		
TC-NFR-03	Planned		
TC-NFR-04	Planned		
TC-NFR-05	Planned		
TC-NFR-06	Planned		
TC-NFR-07	Planned		
TC-NFR-08	Planned		
TC-NFR-09	Planned		
TC-NFR-10	Planned		
TC-NFR-11	Planned		
TC-NFR-12	Planned		
TC-MTR-01	Planned		
TC-MTR-02	Planned		
TC-MTR-03	Planned		
TC-MTR-04	Planned		
SEC-01	Planned		
SEC-02	Planned		

6. SRS Traceability

The traceability below directly reflects the SRS. Every FR, NFR and MTR requirement has the same Test Case ID as listed in the SRS requirement tables/traceability matrix. SR-01 and SR-02 are security validation tests.

SRS Requirement	Test Plan ID	Traceability Area
FR-01	TC-FR-01	Functional requirement validation
FR-02	TC-FR-02	Functional requirement validation

FR-03	TC-FR-03	Functional requirement validation
FR-04	TC-FR-04	Functional requirement validation
FR-05	TC-FR-05	Functional requirement validation
FR-06	TC-FR-06	Functional requirement validation
FR-07	TC-FR-07	Functional requirement validation
FR-08	TC-FR-08	Functional requirement validation
FR-09	TC-FR-09	Functional requirement validation
FR-10	TC-FR-10	Functional requirement validation
FR-11	TC-FR-11	Functional requirement validation
FR-12	TC-FR-12	Functional requirement validation
FR-13	TC-FR-13	Functional requirement validation
FR-14	TC-FR-14	Functional requirement validation
FR-15	TC-FR-15	Functional requirement validation
FR-16	TC-FR-16	Functional requirement validation
FR-17	TC-FR-17	Functional requirement validation
FR-18	TC-FR-18	Functional requirement validation
FR-19	TC-FR-19	Functional requirement validation
FR-20	TC-FR-20	Functional requirement validation
FR-21	TC-FR-21	Functional requirement validation
FR-22	TC-FR-22	Functional requirement validation
FR-23	TC-FR-23	Functional requirement validation
FR-24	TC-FR-24	Functional requirement validation
FR-25	TC-FR-25	Functional requirement validation
NFR-01	TC-NFR-01	Non-functional validation/review
NFR-02	TC-NFR-02	Non-functional validation/review
NFR-03	TC-NFR-03	Non-functional validation/review
NFR-04	TC-NFR-04	Non-functional validation/review
NFR-05	TC-NFR-05	Non-functional validation/review
NFR-06	TC-NFR-06	Non-functional validation/review
NFR-07	TC-NFR-07	Non-functional validation/review
NFR-08	TC-NFR-08	Non-functional validation/review
NFR-09	TC-NFR-09	Non-functional validation/review
NFR-10	TC-NFR-10	Non-functional validation/review
NFR-11	TC-NFR-11	Non-functional validation/review
NFR-12	TC-NFR-12	Non-functional validation/review
MTR-01	TC-MTR-01	Technical verification/review
MTR-02	TC-MTR-02	Technical verification/review
MTR-03	TC-MTR-03	Technical verification/review
MTR-04	TC-MTR-04	Technical verification/review
SR-01	SEC-01	Authentication security validation
SR-02	SEC-02	Role-based access security validation

6.1 Requirement Coverage Summary

Requirement Group	SRS IDs Covered	Test Plan Coverage
Functional Requirements	FR-01 to FR-25	TC-FR-01 to TC-FR-25
Non-Functional Requirements	NFR-01 to NFR-12	TC-NFR-01 to TC-NFR-12
Technical Requirements	MTR-01 to MTR-04	TC-MTR-01 to TC-MTR-04
Security Requirements	SR-01 to SR-02	SEC-01 to SEC-02